



The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR A single **linear map** on residual activations predicts an LLM’s answer from its context **before generation** held-out R^2 0.75, and rank-1 retrieval of the true answer among 9,941 candidates at 0.98, at the fresh-draw ceiling is already $\sim 87\%$ present in the **base model**, is **shared between the assistant and fictional characters**, and predicts **sycophancy, hallucination, and misalignment before inference**.

MOTIVATION

- LLMs are trained as **next-token predictors**, but the weights fix a deterministic mapping from a context to a distribution over *whole answers*: they are also **next-answer predictors**. Absent tool results, **everything about the answer distribution is fixed before the first token is generated**.
- A **simple approximation** of that mapping would be extremely useful: predict behavior *before* generation, and predict how fine-tuning will change it.
- But it need not exist: the mapping is billions of parameters applying dozens of nonlinear layers. Prior work sits at two ends of a **capacity–usefulness tradeoff** cheap but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a whole-distribution imitator with no capacity advantage over the model itself.
- **Our question**: can a **very low-capacity** approximation capture the mapping *as a whole*? Such an approximation is a **metamodel**: a much smaller model that reads the LLM’s internals and predicts its behavior.

THE RESIDUAL CONTEXT-ANSWER MAPPING

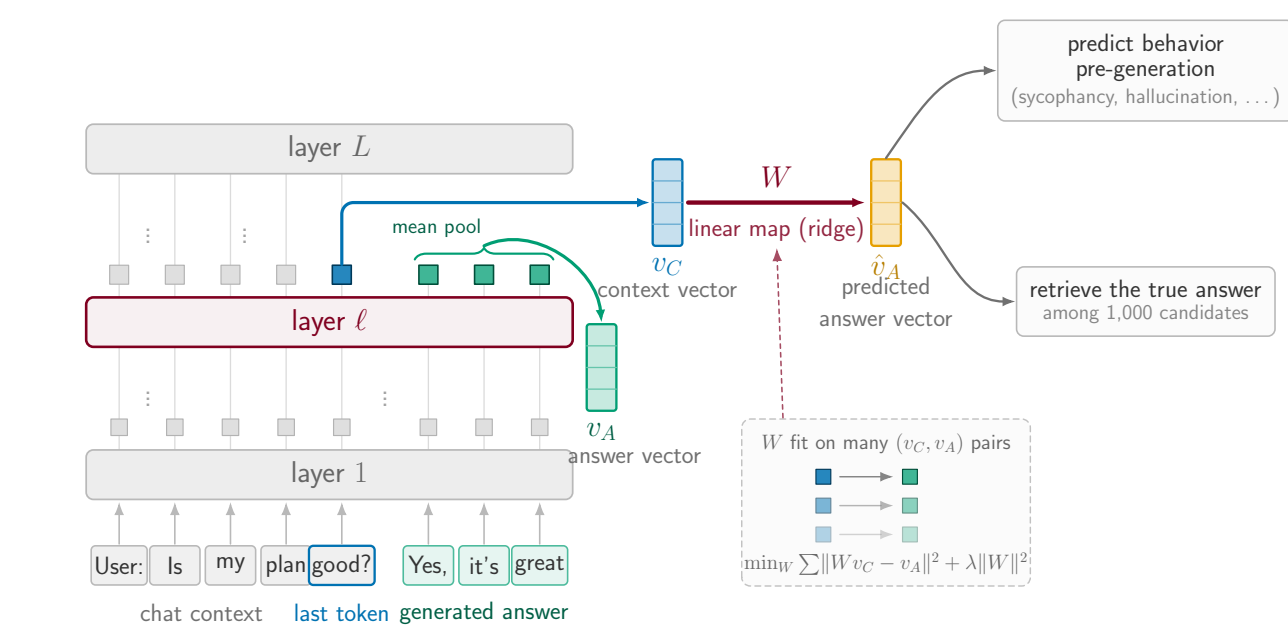


Figure 1. The **context vector** v_C is the residual-stream state at the last context token (layer ℓ); the **answer vector** v_A is the mean answer-token state. Both summaries follow prior work. *Fitted maps* $v_C \rightarrow v_A$ approximate the mapping; readouts consume v_A .

EXPERIMENTS

Qwen2.5-7B(-Instruct) headline; Llama/Tülu post-training ladder. Real-user contexts (LMSYS-Chat-1M, WildChat), up to 963k pairs. Ridge maps per layer, cross-validated λ . Behaviors LLM-judged. Retrieval is reported raw-euclidean unless stated; the whitened-cosine + CSLS read is strictly higher.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

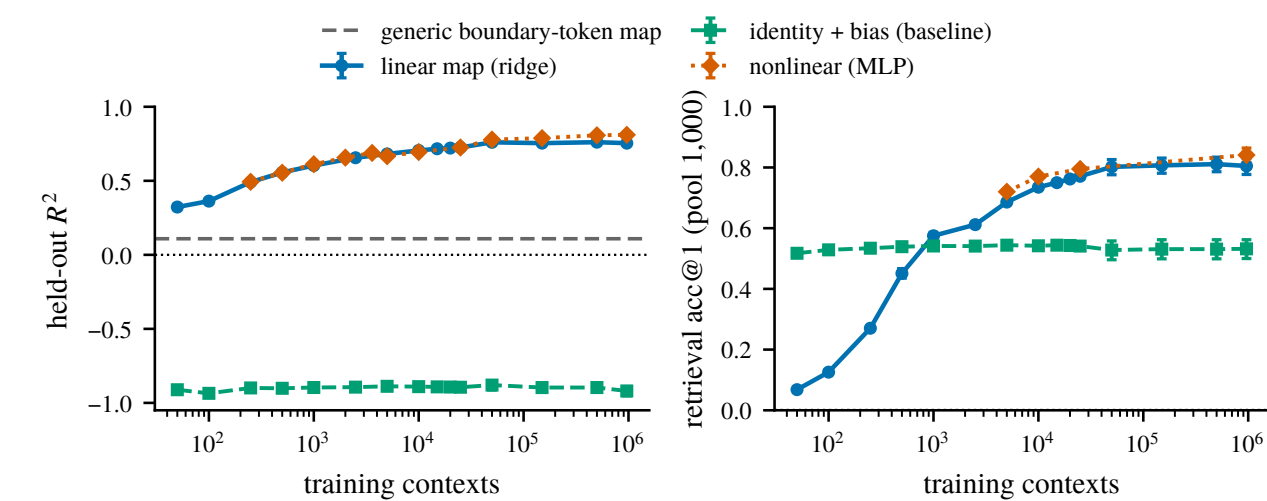


Figure 2. Held-out R^2 (left) and rank-1 retrieval (right, *raw euclidean*, pool 1,000) against training-set size, for the linear map, a nonlinear (MLP) map, and the identity + bias baseline. Dashed: a generic boundary-token map. Qwen2.5-7B-Instruct, layer 19. Raw euclidean understates retrieval: on the multi-turn map, a whitened-cosine + CSLS read lifts rank-1 from 0.82 to 0.98 (pool 9,941).

- **High predictive power**: held-out R^2 0.75 at 963k training contexts, and rank-1 retrieval far above chance (10^{-3}) at every training size.
- **Mostly linear**: the nonlinear (MLP) map adds only ~ 0.06 R^2 over the linear map, and only at large n .
- **Far above the baselines**: identity + bias never leaves the negative- R^2 regime; a generic boundary-token (punctuation) $\rightarrow$ next-segment map reads $R^2 \approx 0.11$ (dashed) so this is not just residual-stream smoothness.
- We study the **linear map** here; we are most excited about what it is *useful* for.

2. IT PREDICTS THE PERSONA-VECTOR DIRECTIONS

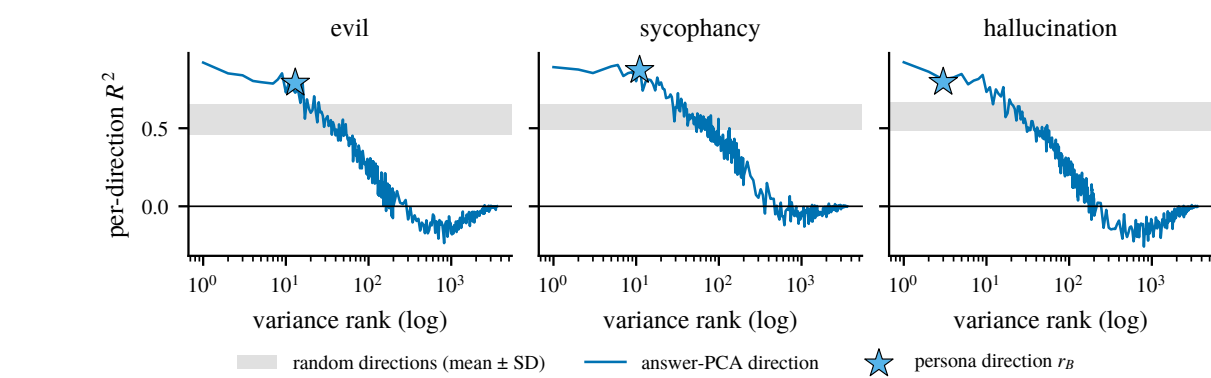


Figure 3. Per-direction held-out R^2 along answer-PCA directions against variance rank, per behavior. Star: the persona direction r_B at its equivalent variance rank. Gray band: mean ± 1 SD over 50 random unit directions.

- **Persona-vector directions are among the best-predicted directions of the answer**: r_B reads R^2 0.79 / 0.87 / 0.80 (misalignment / sycophancy / hallucination) against a random-direction band of ~ 0.56 –0.58.
- **They sit at the very top of the answer’s variance spectrum** (equivalent rank 2–12 of 3,584), where per-direction R^2 is highest; predictability decays with rank and crosses zero past ~ 160 –360.

3. BEST AT HIGH-LEVEL, PERSONA-LIKE PROPERTIES

To ask *what* it predicts, we fit maps to the answer’s **SAE features** (16,384 features; per-feature held-out R^2) and rank feature properties by how well they predict a feature’s R^2 .

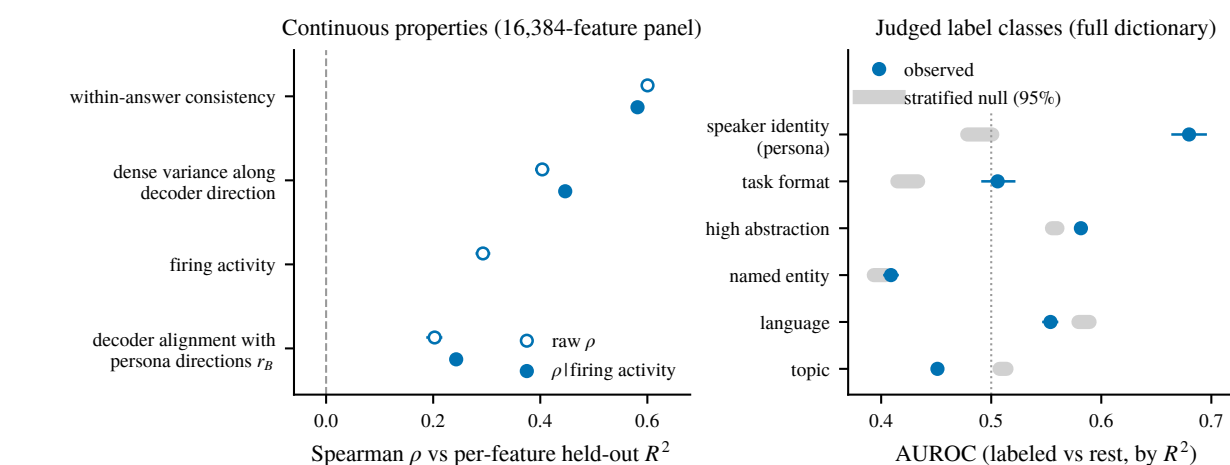


Figure 4. Which properties of an SAE feature predict its per-feature R^2 , on a common footing against matched nulls.

- **Persona / speaker-identity features are the best-predicted label class**; content entities and token-level word choice are the worst.
- **Predictability follows abstraction**: median per-feature R^2 falls $0.43 \rightarrow 0.04$ from general to specific features.
- **Within-answer consistency is the strongest single correlate** tonic, context-determined properties (language, register, discourse position) predict well; phasic token-level choices do not.

4. WHAT DOES IT FAIL ON?

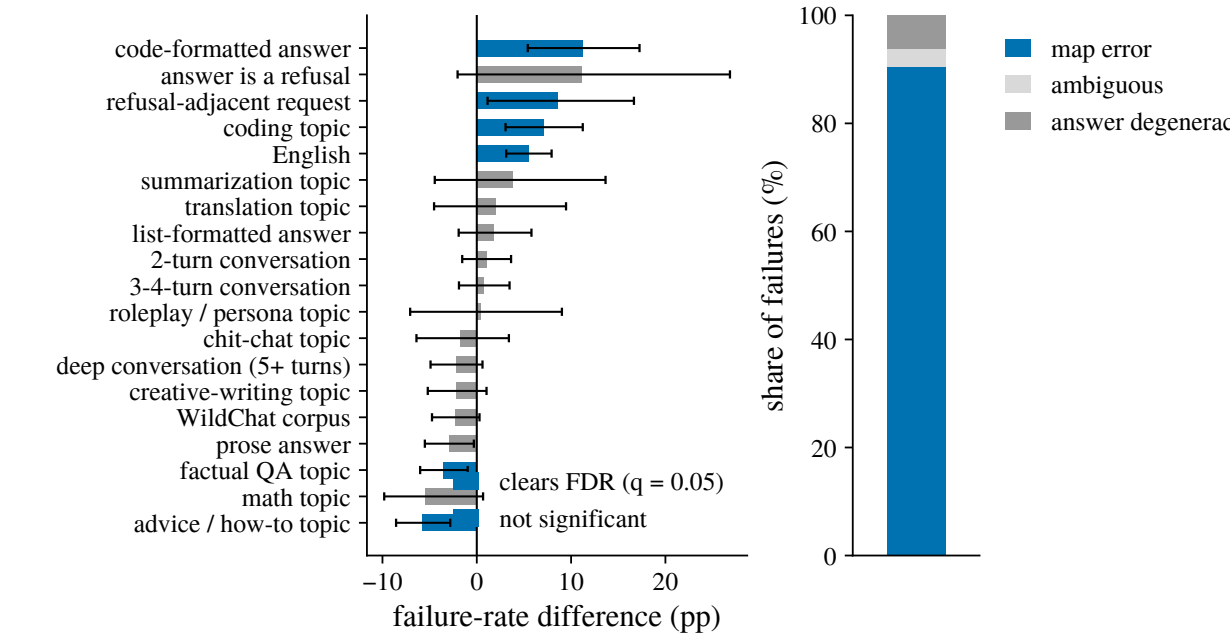


Figure 5. Failures against *sampling-noise-free* answer targets (each answer averaged over 5 on-policy draws); raw-euclidean retrieval, $n = 1,988$ resample-covered held-out contexts, pool 9,941, 180 failures, rank-1 accuracy 0.909. Left: failure-rate difference by context category (pp); blue clears Benjamini–Hochberg FDR at $q = 0.05$, gray tested but not significant. Right: attribution of the 180 failures (100% covered); *map error* = the answer is still retrievable draw-vs-draw ($s \geq 0.75$), so the miss is the map’s.

- **Failures concentrate in identifiable categories**: code-formatted answers (+11.2pp), refusal-adjacent requests (+8.6), coding topics (+7.0), English (+5.6); factual QA (−3.6) and advice/how-to (−5.8) fail least.
- **Almost entirely map error, not answer degeneracy**: 91% of rank-1 misses stay retrievable from a fresh draw of the same answer; only 6% are

This work was carried out as part of the MATS program.

5. WHERE DOES THE MAPPING COME FROM?

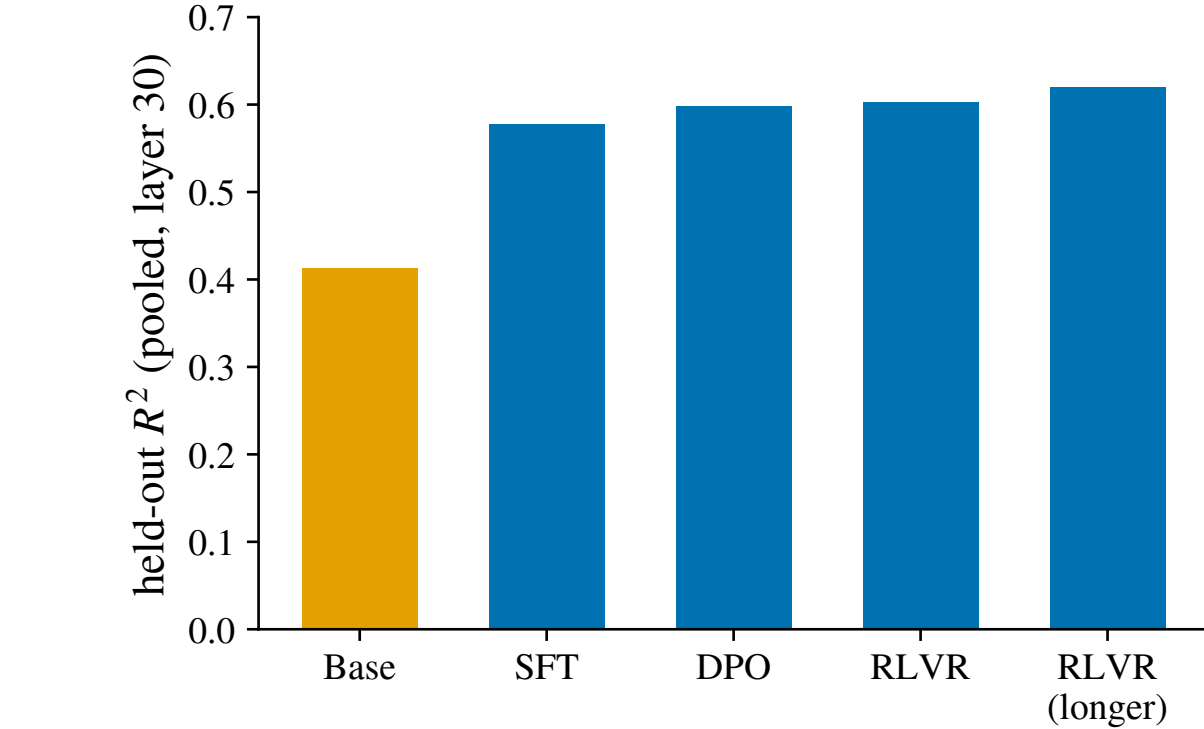


Figure 6. Pooled held-out R^2 of the fitted map at each stage of the open Llama/Tülu post-training ladder (layer 30), each stage fit and evaluated on its own on-policy answers.

- **Already largely present in the base model**: the base bar reads R^2 0.41 before any post-training. On Qwen, with corpus, answer text, and recipe held identical, base reaches $\sim 87\%$ of instruct (0.588 vs. 0.673).
- **SFT is by far the largest step** (+0.17), and **DPO $\rightarrow$ RLVR barely moves it** (+0.004) post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

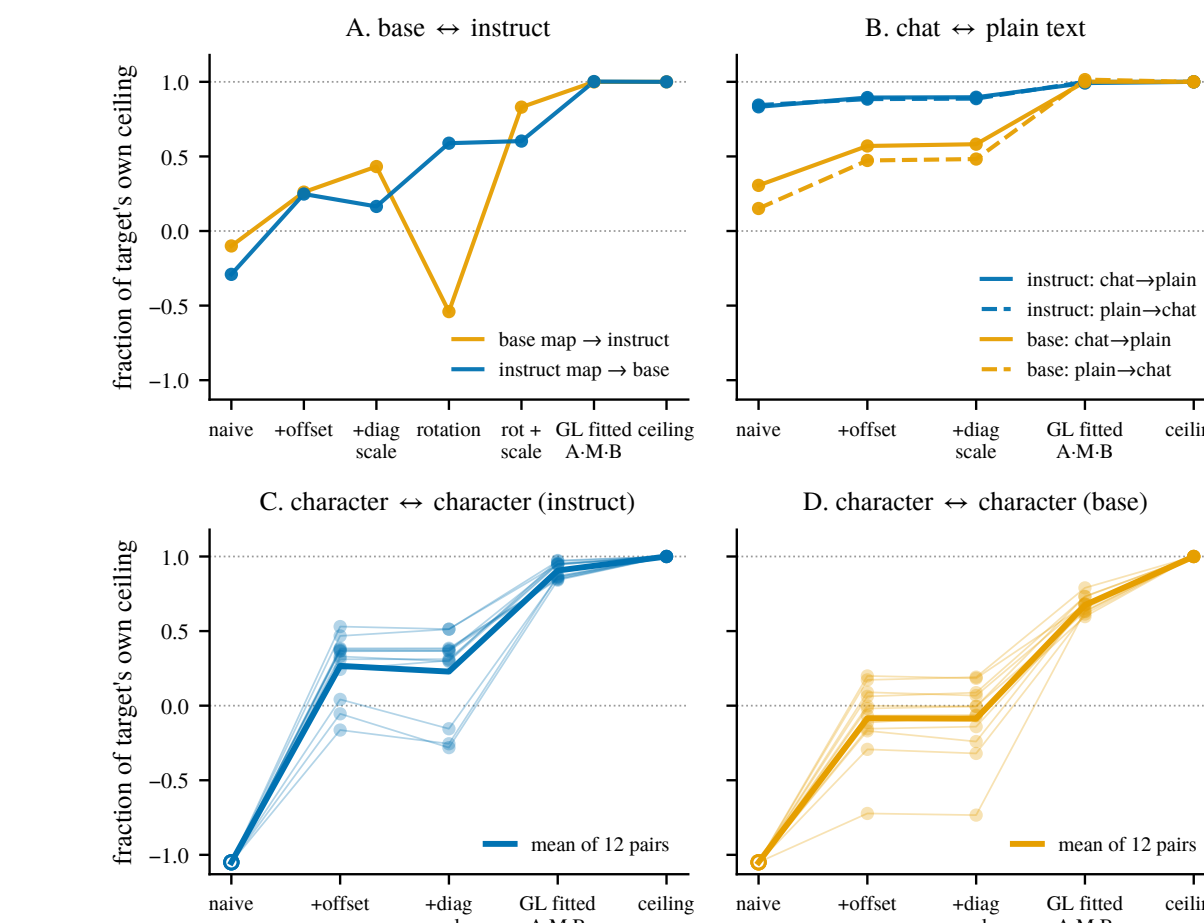


Figure 7. Fraction of the target’s own held-out ceiling recovered when one fitted map is applied to another condition, across correction rungs (naive $\rightarrow$ +offset $\rightarrow$ +diag. scale $\rightarrow$ fitted linear reparameterization $\rightarrow$ ceiling).

- **One shared operator serves the assistant and fictional characters**: character $\leftrightarrow$ character maps reach 84–97% of the target’s own ceiling once a *linear reparameterization* is fitted (C, D); naive transfer fails, so sharing is up to a change of coordinates.
- **The same ladder closes across framings and post-training** (mean plain-text misalignment; assistant is one character on a shared map.

IS THIS USEFUL? BEHAVIOR PREDICTION BEFORE INFERENCE

- **Persona-vector readouts on the mapped answer track judged sycophancy, hallucination, and misalignment near the real-answer ceiling** (misalignment: ρ 0.67 vs. 0.69 reading the real answer), while the same readout on the raw context collapses off-synthetic prompts ($\rho \leq 0.03$).
- **Why not just probe the context? The map exploits unlabeled data**: fit on in-domain *unjudged* text, mapped-answer probing beats context probing at every label budget ($\sim 3\times$ at 250 labels).

LIMITATIONS

- **Prediction $\neq$ control**: the fitted map’s read direction does not steer behavior, while the mean-difference persona vector does.
- R^2 and retrieval dissociate across estimators, so map-strength claims are metric-specific; headline results are per model family and correlational.

IMPLICATIONS & FUTURE WORK

- Much of an LLM’s behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool, and evidence for the **persona selection model**.
- Next: a dynamical-systems account of the map; predicting answer *correctness*; post-training as “making the answer more predictable from the context”; automated redteaming via the map’s pre-image.

REFERENCES

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